

Shashank Yadav Kalki

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PROFESSIONAL SUMMARY

Software Engineer with a strong foundation in object-oriented programming, RESTful API design, microservices, and scalable backend systems, complemented by a Machine Learning postgraduate background in AI and Generative AI applications. Experienced in building end-to-end software pipelines, production-style APIs, and data-driven systems, with a focus on reliability, performance, and maintainability. Interested in designing and developing high-resiliency, customer-facing systems for large-scale enterprise environments.

EDUCATION

University College Cork	Cork, Ireland
M.Sc. Mathematical Modelling & Machine Learning (Second Class Honours Grade I)	2024 Sept – 2026 March
Mahatma Gandhi Institute of Technology	Hyderabad, India
B.Tech Computer Science (Data Science)	2020 – 2024

TECHNICAL SKILLS

Programming Languages: Python (NumPy, pandas, scikit-learn), Java, R
Software Engineering: OOP Design, RESTful APIs, Microservices, SOA, Design Patterns
Backend Frameworks: Django, FastAPI, Flask **Databases:** PostgreSQL, Oracle DB, Cassandra, SQL
DevOps & CI/CD: Docker Git/GitHub
Systems & Networking: Linux OS, Command-line Tools, TCP/IP, DNS, HTTP
Machine Learning & AI: Regression, Classification, Deep Learning (CNN, RNN/LSTM), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, OpenAI API

PROJECTS

Confabulation Dynamics in Reservoir Computing – Research Project	May 2025 – Sept 2025
<i>University College Cork Under Dr. Andrew Flynn</i>	
- Studied how reservoir computing systems generate “confabulations” (plausible but incorrect outputs). - Analyzed dynamical systems behavior and emergence of untrained attractors. - Investigated model reliability and failure modes in recurrent neural networks. - Connected concepts from nonlinear dynamics with machine learning system behavior.	
Document Intelligence Microservices Platform	Jan 2025 – Feb 2025
<i>Python FastAPI Microservices Docker OpenAI API ChromaDB</i>	
- Architected a scalable, microservices-based platform for extracting, summarizing, and querying information from unstructured documents. - Designed and exposed RESTful APIs using FastAPI, following OOP and SOA principles to enable clean integration with external applications. - Built modular document ingestion and embedding pipelines, separating concerns across independently deployable services. - Containerized services with Docker to support consistent deployment, horizontal scalability, and environment parity. - Implemented input validation, structured logging, and error handling to improve system reliability and observability.	
Multi-Agent Research Assistant	Aug 2025 – Sept 2025
<i>Python LangGraph LangChain OpenAI API ChromaDB</i>	
- Developed a multi-agent AI system capable of coordinating specialized agents for retrieval, summarization, and response generation tasks. - Implemented agent orchestration workflows using LangGraph to enable task decomposition and collaboration between agents. - Integrated tool-calling capabilities allowing agents to access external knowledge sources and APIs. - Designed short-term memory mechanisms to maintain context across multi-step interactions. - Evaluated system performance for response quality, task completion, and reliability across diverse research queries.	
Scalable Knowledge Retrieval Backend Service	Sept 2024 – Nov 2024
<i>Python FastAPI PostgreSQL OpenAI API ChromaDB Docker</i>	
- Built an end-to-end backend service exposing RESTful APIs for semantic search and retrieval across 10,000+ documents. - Designed a hybrid data architecture combining PostgreSQL with a vector store, reducing retrieval latency by 30%. - Structured the codebase into modular, service-oriented components following OOP and SOA principles for scalability and maintainability. - Integrated OpenAI APIs with FastAPI microservices to deliver grounded, context-aware responses under production-like load. - Used Git for version control and Docker for containerized deployment, ensuring consistent environments across development stages.	

CERTIFICATIONS

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| <ul style="list-style-type: none">Data Science for Engineers – IIT MadrasNatural Language Processing – IIT KharagpurData Analysis with Python – FreeCodeCamp | <ul style="list-style-type: none">Social Network Analysis – IIT RoparPrompt Engineering with OpenAI – Columbia UniversityGenerative AI Masterclass – Udemy |
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